

Electrochemically Controlled Synthesis of Ultrathin Nickel Hydroxide Nanosheets for Electrocatalytic Oxygen Evolution

Li-Ming Cao, Qing-Cai Cao, Jia Zhang, Xuan-Yi Zhu, Rong-Zhi Sun, Zi-Yi Du, and Chun-Ting He*

Cite This: *Inorg. Chem.* 2021, 60, 3365–3374

Read Online

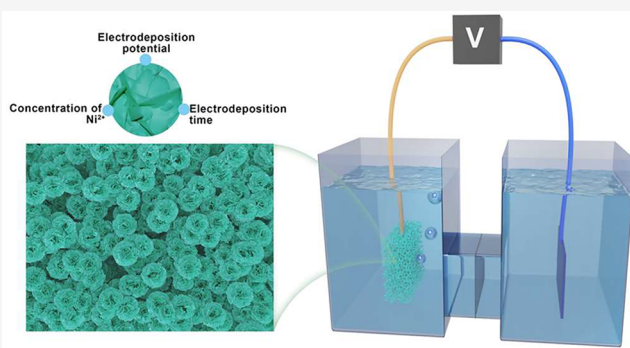
ACCESS |

Metrics & More

Article Recommendations

Supporting Information

ABSTRACT: The development of oxygen evolution reaction (OER) catalysts with high activity and high stability through convenient and economical methods is greatly important for the promotion of hydrogen energy based on electrolysis technology. Herein, by using an unconventional high electrodeposition potential, novel petal-like clusters constructed by cross-linking ultrathin nickel hydroxide nanosheets were controllably synthesized on nickel foam (or copper foam or carbon cloth) and the effect of electrodeposition conditions on their OER performance was carefully explored. Due to the abundant catalytically active sites, promoting electron conduction/mass transmission from the specific micro–nano structure, as well as the ultrasmall thickness of ~ 3.0 nm, the optimized α -Ni(OH)₂/NF self-supporting electrode exhibits excellent electrocatalytic performance for OER, merely requiring low overpotentials of 192 and 240 mV to yield current densities of 10 and 100 mA cm⁻² in 1.0 M KOH, respectively, which surpassed those of all of the reported nickel hydroxide/oxides and the benchmark RuO₂. Moreover, α -Ni(OH)₂/NF can drive the high-current density (500–1000 mA cm⁻²) OER at low overpotentials, meeting the requirements of potential industrial applications.



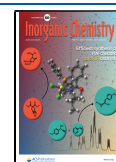
1. INTRODUCTION

The depletion of fossil fuels and growing environmental problems have forced people to seek sustainable and clean energy.^{1,2} Hydrogen (H₂) has the advantages of a high combustion value and no pollution, making it an ideal source of renewable clean energy.^{3–5} The electrolysis of water to produce H₂ not only is simple and practical but also can yield high-purity H₂, which is a promising approach for the conversion of renewable energy into hydrogen energy.^{6–8} Water splitting is composed of two key half-reactions: oxygen evolution reaction (OER) and hydrogen evolution reaction (HER).^{9–11} Between them, HER has great thermodynamic advantages.^{12,13} In contrast, OER involves a complex multi-proton coupled electron transfer process, resulting in slower oxygen evolution kinetics, which requires a higher overpotential to accelerate the reaction, thus becoming the bottleneck of water splitting technology.^{14–18} Therefore, it is very important to prepare highly active OER electrocatalysts to decrease the activation energy of the reaction and accelerate the kinetics of oxygen evolution, thereby improving the efficiency of water splitting. Currently, Ir/Ru oxides are recognized as one kind of the most active OER electrocatalysts.¹⁹ However, the scarcity and high cost of these precious metal catalysts have seriously hindered their large-scale commercial application. As a consequence, it is particularly important to develop low-cost, available, and highly active non-noble metal catalysts, such as earth-abundant

transition metal (Fe, Co, Ni, Mn, etc.)-based electrocatalysts.^{20–23} In particular, cobalt-based electrocatalysts, such as Co-based oxides/hydroxides, sulfides, selenides, phosphides, and borides, have been widely developed for their remarkable OER performance.^{24–26} For example, Wang et al. synthesized ultrathin CoFe LDH nanosheets by a hydrothermal method, and the catalyst needed an overpotential of only 233 mV to yield a current density of 10 mA cm⁻² in 1.0 M KOH.²⁴ Compared with cobalt, nickel usually exhibits much weaker OER activity, although it is much more abundant and cheaper.²⁷ For instance, the OER performances of pure nickel oxide/hydroxide electrocatalysts are generally rather undesirable.^{28–33} Han et al. synthesized nickel boride-hydroxide nanoparticles [F-Ni₂B/Ni(OH)_x] by a one-step wet-chemical method, which required an overpotential of 340 mV to yield a current density of 10 mA cm⁻² in 1.0 M KOH.²⁸ Liu et al. synthesized Ni(OH)₂ nanosheets with a thickness in the range of 20–100 nm by a hydrothermal method, requiring an overpotential of 308 mV to yield a current density of 10 mA

Received: December 23, 2020

Published: February 11, 2021



cm^{-2} in 1.0 M KOH.³⁰ Tian et al. reported that NiOOH nanosheets with a thickness of 30 nm required an overpotential of 266 mV to yield a current density of 10 mA cm^{-2} .³¹ Undoubtedly, improving the performance of nickel-based catalysts would be advantageous for the actual industrial application.

Electrodeposition is a convenient and universal technology that can be used for large-scale industrial production.^{34,35} It needs simple and low-cost equipment to precisely control the nucleation and growth of substances to prepare electroplated products with different structures and morphologies.^{36,37} Direct electrodeposition of electrochemically active species on conductive substrates can construct self-supporting electrode materials for supercapacitors, batteries, and electrocatalysis.^{38–41} Compared with the powdered catalysts, the electrodeposited self-supporting electrodes, on one hand, can avoid the use of expensive conductive polymer adhesives and possess quite abundant and highly exposed electrochemically active sites; on the contrary, they show excellent stability, especially working under high current densities, in which a large number of O_2 bubbles are easy to peel off the glued powder catalysts, yet the catalysts directly grown on the conductive substrate are more robust.^{42,43} Currently, there have been some reports of electrodeposited nickel hydroxide/nickel oxide self-supporting electrodes for OER, but they are mostly thick nanoplatelets or nanoparticles with unsatisfactory catalytic activities.^{44–47} For example, Ni et al. prepared nickel hydroxide nanoparticles with a size of $\sim 20 \text{ nm}$ on nickel foam (NF) by electrodeposition, which required an overpotential of 366 mV to yield a current density of 100 mA cm^{-2} in 1.0 M KOH.⁴⁴ Geng et al. synthesized $\alpha\text{-Ni(OH)}_2$ nanosheets on a NF/N-CNT substrate via electrodeposition under a constant cathodic current of 2 mA cm^{-2} , which required an overpotential of 254 mV to yield a current density of 10 mA cm^{-2} in 1.0 M KOH.⁴⁷ In addition, most of the electrodeposition potentials ever used in the synthesis of nickel hydroxides/nickel oxides do not exceed -1.0 V versus Ag/AgCl.^{44–47} In fact, the electrodeposition potential plays an important role in the morphology and structure of nanomaterials, and under a high potential or a high current density, the ultrafast ion migrations may lead to special micro–nano structures and/or novel active sites.⁴¹ In addition, the other electrodeposition conditions, such as electrodeposition time, electrolyte concentrations, and anion species, are also significant for the electrochemical in situ synthesis,⁴¹ yet their specific effects on the morphology and structure of nickel hydroxides are rarely studied.

Herein, we synthesized three-dimensional (3D) petal-like clusters constructed by cross-linking two-dimensional (2D) ultrathin nickel hydroxide nanosheets on NF by using a relatively high electrodeposition potential and carefully explored the effect of electrodeposition conditions on the OER performance of nickel hydroxide. We found that the electrodeposition potential strongly influences the formation of petal-like clusters, thereby affecting the catalytic performance of nickel hydroxide. Benefiting from the novel micro–nano structure as well as the rather small thickness of $\sim 3.0 \text{ nm}$, the obtained nickel hydroxide nanosheets possess abundant catalytically active sites, promoting electron conduction/mass transmission, and remarkable stability. Therefore, the electrodeposited $\alpha\text{-Ni(OH)}_2$ /NF self-supporting electrode prepared under optimal parameters exhibits excellent electrocatalytic performance for OER, merely requiring low overpotentials of

192, 240, and 290 mV to yield current densities of 10, 100, and 250 mA cm^{-2} in 1.0 M KOH, respectively. Moreover, $\alpha\text{-Ni(OH)}_2$ /NF can even drive the high current density ($500\text{--}1000 \text{ mA cm}^{-2}$) at low overpotentials, indicating that it has potential industrialization prospects.

2. EXPERIMENTAL SECTION

2.1. Synthesis of Materials. For the synthesis of $\alpha\text{-Ni(OH)}_2$ nanosheets on nickel foam (NF, thickness of 1.7 mm), NF was ultrasonically cleaned in ethanol, a 3.0 M HCl solution, and ultrapure water for 10 min. To fabricate $\alpha\text{-Ni(OH)}_2$ /NF, electrodeposition was performed at different electrodeposition potentials employing a three-electrode system with a NF working electrode, a Pt counter electrode, and a Ag/AgCl (in a 3 M KCl solution) reference electrode. An aqueous solution of $\text{Ni(NO}_3)_2 \cdot 6\text{H}_2\text{O}$ was used as the electrolyte. The NF working electrode ($1 \text{ cm} \times 3 \text{ cm}$) was immersed in the electrolyte. The detailed experimental process is as follows (take the electrodeposition conditions with the deposition potential of -2.0 V vs Ag/AgCl, an electrolyte concentration of 0.2 M, and a deposition time of 300 s as the example). First, a 0.2 M $\text{Ni(NO}_3)_2 \cdot 6\text{H}_2\text{O}$ aqueous solution was poured into the electrolytic cell. Then, the NF working electrode (immersion area of $1 \text{ cm} \times 3 \text{ cm}$), a Pt counter electrode, and a Ag/AgCl (in a 3 M KCl solution) reference electrode were inserted into the electrolyte and then applied to a cathodic deposition at -2.0 V for 300 s. To fabricate $\alpha\text{-Ni(OH)}_2$ /NF at different electrodeposition potentials, it is only necessary to set the cathode deposition potentials to -0.1 , -1.5 , -2.5 , or 3.0 V , and the other synthesis conditions are the same. To fabricate $\alpha\text{-Ni(OH)}_2$ /NF at different electrolyte concentrations, it is only necessary to change the electrolyte concentration to 0.1, 0.3, or 0.4 M, and the other synthesis conditions are the same. To fabricate $\alpha\text{-Ni(OH)}_2$ /NF at different electrodeposition times, it is only necessary to change the electrodeposition time to 200, 400, or 500 s, and the other synthesis conditions are the same. To fabricate $\alpha\text{-Ni(OH)}_2$ /CF and $\alpha\text{-Ni(OH)}_2$ /CC, copper foam and carbon cloth are used as the working electrodes, respectively, and the other synthesis conditions are the same.

2.2. Characterization. Scanning electron microscopy (SEM, ZEISS EVO 10) and transmission electron microscopy (TEM, JEM2100F) were utilized to characterize the morphology of the catalyst. The D8 ADVANCE X-ray diffractometer was used to obtain the powder X-ray diffraction (PXRD) data. The ESCA Lab250 X-ray photoelectron spectrometer was used to obtain the X-ray photoelectron spectroscopy (XPS) data.

2.3. Electrochemical Measurements. Electrochemical measurements were performed in 1.0 M KOH through a model CHI760E electrochemical workstation employing $\alpha\text{-Ni(OH)}_2$ /NF, Hg/HgO, and Pt plate as the working, reference, and counter electrodes, respectively. The loading mass of $\alpha\text{-Ni(OH)}_2$ on the substrate is 9.6 mg cm^{-2} . For comparison, we loaded commercial RuO_2 with the same mass on NF. The homogeneous catalyst ink was prepared by dispersing RuO_2 (25 mg) in ethanol ($450 \mu\text{L}$) and a 5% Nafion solution ($50 \mu\text{L}$) with ultrasonication. Subsequently, the as-obtained ink ($192 \mu\text{L}$) was evenly spread on a piece of clean NF. The as-prepared $\alpha\text{-Ni(OH)}_2$ /NF self-supporting electrode was directly utilized as the working electrode. All electrodes were activated to a stable state through cyclic voltammetry (CV) before their electrochemical performance was tested. CV activation was carried out in the sweep range of $1.1\text{--}1.8 \text{ V}$ versus a reversible hydrogen electrode (RHE) with a scan rate of 100 mV s^{-1} . The electrochemical surface area (ECSA) was evaluated by measuring the electrochemical double-layer capacitance (C_{dl}), which was measured by CV measurements in the non-Faradaic region ($0.93\text{--}1.03 \text{ V}$ vs RHE) at scan rates of 60, 80, 100, 120, 140, and 160 mV s^{-1} . A linear relationship between the capacitive currents at 0.98 V versus RHE with the scan rates could be obtained. The C_{dl} value was calculated, which is half of the linear slope. Electrochemical impedance spectroscopy (EIS) was utilized to measure the charge transfer resistance (R_{CT}) of the catalyst, which was carried out with a frequency ranging from 0.01 Hz to 100 kHz. $\alpha\text{-}$

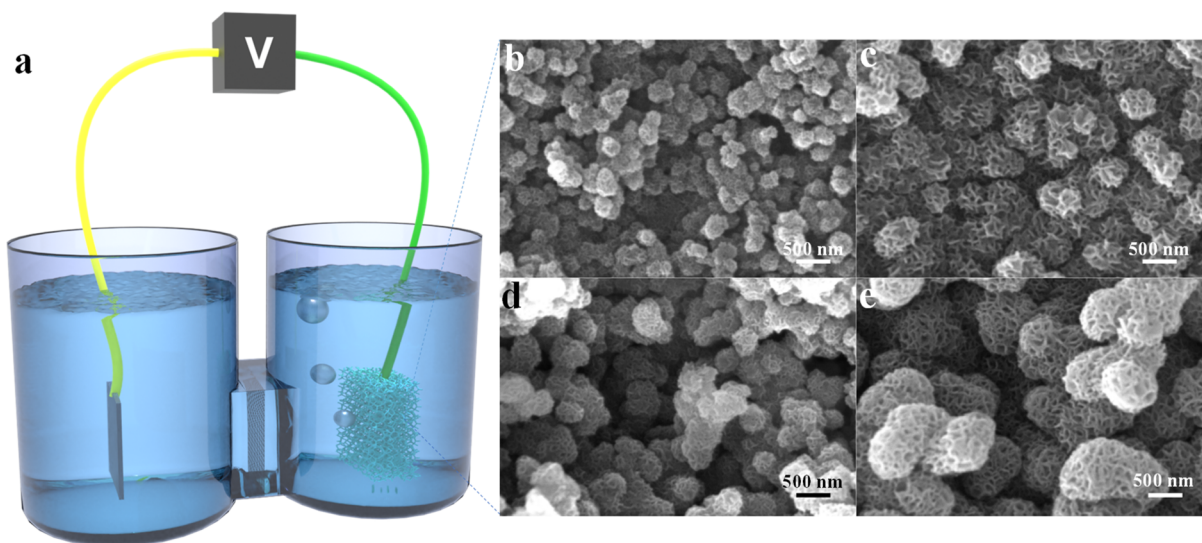


Figure 1. (a) Schematic illustration of the electrochemically controlled synthesis of the α -Ni(OH) $_2$ /NF self-supporting electrode. SEM images of α -Ni(OH) $_2$ obtained at different electrodeposition potentials (b) -1.0 V vs Ag/AgCl, (c) -2.0 V vs Ag/AgCl, (d) -2.5 V vs Ag/AgCl, and (e) -3.0 V vs Ag/AgCl.

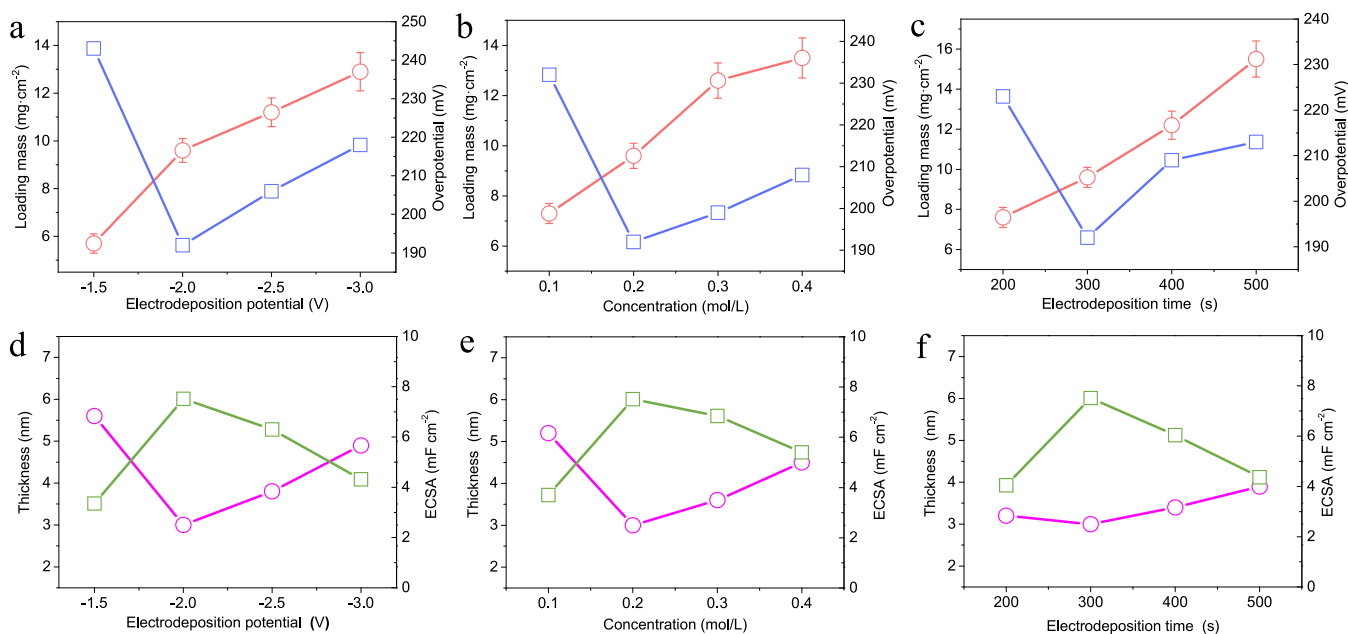


Figure 2. Loading masses (red circles), overpotentials at 10 mA cm^{-2} in 1.0 M KOH (blue squares), nanosheet thicknesses (magenta circles), and ECSA (green squares) of α -Ni(OH) $_2$ /NF obtained at different (a and d) electrodeposition potentials, (b and e) $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ concentrations, and (c and f) electrodeposition times.

Ni(OH) $_2$ /NF was used as the working electrode to measure the Faraday efficiency for OER. Galvanostatic electrolysis measurement was carried out at a current density of 10 mA cm^{-2} for 3 h. The gas produced was analyzed on a gas chromatograph. The TOF value was calculated with the equation $\text{TOF} = j/(4Fn)$, where j is the current density (milliamperes per square centimeter) at a given overpotential, F is the faraday constant (96485 C mol^{-1}), and n is the number of moles of the electrocatalyst (moles per square centimeter). All of the catalytically related metal atoms were assumed to be accessible for catalysis of OER.

3. RESULTS AND DISCUSSION

The commercial NF with excellent electrical conductivity was selected as the substrate to fabricate the α -Ni(OH) $_2$ /NF self-supporting electrode (Figure 1a). α -Ni(OH) $_2$ /NF was

prepared by a simple and convenient electrodeposition method. During the preparation process, the nitrate ions in the $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ electrolyte migrated to the vicinity of the NF electrode under the action of an external electric field and were reduced to produce OH^- .⁴¹ Deposition could occur when the precipitation equilibrium constant of Ni(OH) $_2$ was reached. Therefore, the synthesis mechanism can be expressed as follows: $\text{NO}_3^- + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^- + \text{NO}_2^-$ (cathodic reduction) or $\text{Ni}^{2+} + 2\text{OH}^- \rightarrow \text{Ni}(\text{OH})_2$ (precipitation). To explore the optimal preparation conditions for α -Ni(OH) $_2$ /NF with high electrocatalytic performance, we investigated the effects of electrodeposition potential, electrodeposition time, and electrolyte concentration on the OER performance of the catalysts. The linear sweep voltammetry (LSV) curves (Figure

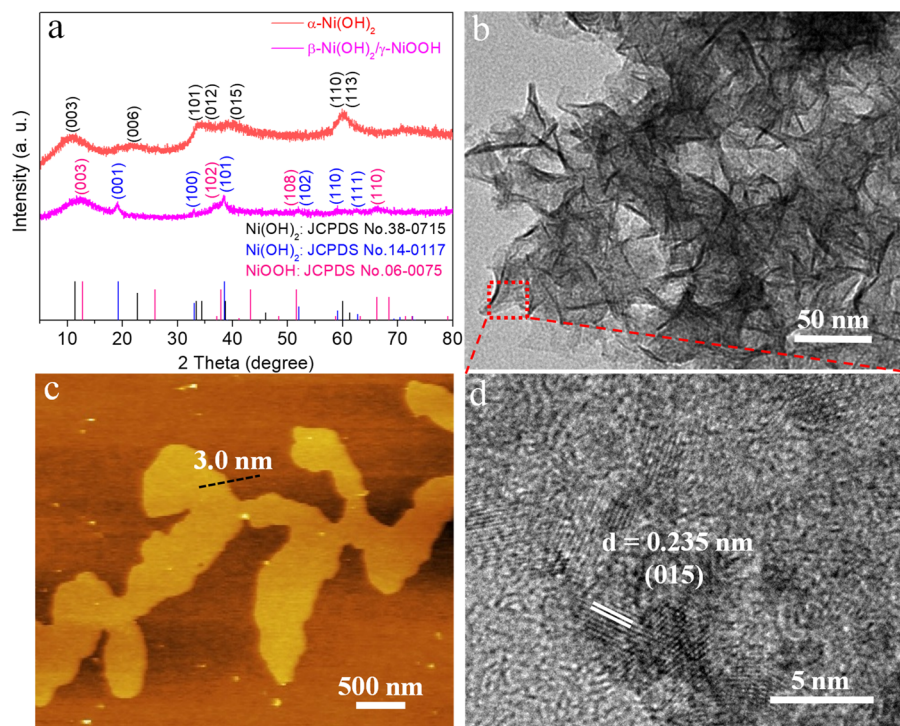


Figure 3. (a) PXRD patterns of α -Ni(OH)₂ scraped from the NF before (up) and after (down) OER. (b) TEM, (c) AFM, and (d) HRTEM images of α -Ni(OH)₂.

S1) and galvanostatic electrolysis measurements at a current density of 10 mA cm⁻² (Figures S2–S4) were employed to evaluate the OER performance of these α -Ni(OH)₂/NF electrodes. First, the effect of electrodeposition potential was studied, and it was found that the loading mass of α -Ni(OH)₂ increases with electrodeposition potential (Figure 2a). The loading mass of α -Ni(OH)₂ greatly increases from -1.5 to -2.0 V versus Ag/AgCl and increases slightly from -2.0 to -3.0 V versus Ag/AgCl. Galvanostatic electrolysis shows that when the loading mass of α -Ni(OH)₂ increased from 5.7 to 9.6 mg, the OER overpotential of the self-supporting electrode decreased significantly, but a continuing increase in the loading would increase the overpotential (Figure 2a and Figure S2). Therefore, the α -Ni(OH)₂/NF prepared by electrodeposition at -2.0 V versus Ag/AgCl has the best OER performance, which demands only an extremely low overpotential of 192 mV to yield a current density of 10 mA cm⁻² in 1.0 M KOH (Figure 2a and Figure S2). Subsequently, we explored the effect of electrolyte concentration on OER performance. The results show that the higher the concentration of Ni(NO₃)₂·6H₂O, the greater the loading mass of α -Ni(OH)₂. The loading mass of α -Ni(OH)₂ greatly increases from 0.1 to 0.3 M and increases slightly from 0.3 to 0.4 M (Figure 2b). Galvanostatic electrolysis reveals that α -Ni(OH)₂/NF fabricated by electrodeposition at 0.2 M Ni(NO₃)₂·6H₂O exhibits the best OER activity (Figure 2b and Figure S3). An overly low or high a loading mass will lead to an increase in OER overpotential. When the loading mass of the catalyst is too low, too few catalytic active sites are provided, which would weaken its catalytic performance for OER. When the loading mass of the catalyst is too high, although it can provide abundant catalytic active sites, it would also hinder mass transfer and charge transfer, thereby weakening its OER catalytic activity. Finally, the effect of electrodeposition time on OER perform-

ance was also investigated. As shown in Figure 2c, the loading mass of α -Ni(OH)₂ almost linearly increases with electrodeposition time. Galvanostatic electrolysis reveals that α -Ni(OH)₂/NF obtained by electrodeposition at 300 s possesses the best OER activity (Figure 2c and Figure S4). An overly low or high loading mass can decrease the catalytic activity. The change in electrodeposition conditions would directly regulate the loading mass of α -Ni(OH)₂, which plays a significant role in boosting the catalytic activity of the self-supporting electrode.

PXRD was used to analyze the composition and crystal phase characteristics of the as-synthesized α -Ni(OH)₂. As displayed in Figure 3a and Figure S5, all of the samples from different electrochemical conditions exhibited almost the same PXRD patterns. The diffraction peaks at 11.32°, 22.73°, 33.46°, 34.41°, 38.76°, 59.98°, and 61.25° can be attributed to the (003), (006), (101), (012), (015), (110), and (113) crystal planes of Ni(OH)₂·0.75H₂O, respectively (JCPDS Card No. 38-0715), indicating the generation of crystalline α -Ni(OH)₂. SEM was utilized to characterize the morphology of the self-supporting electrodes. We found that the electrodeposition potential has the most significant effect on the morphology of α -Ni(OH)₂. At an electrodeposition potential of -1.0 V versus Ag/AgCl, α -Ni(OH)₂ nanosheets grew directly on the surface of NF without forming petal-like clusters (Figure 1b and Figure S6a). When the electrodeposition potential is higher than -1.2 V versus Ag/AgCl, these 2D nanosheets are cross-linked with each other to construct 3D petal-like clusters, which grow uniformly on the entire surface of the NF framework and form a loose and porous coating (Figure 1c–e and Figure S6c–f). Interestingly, we found that as the electrodeposition potential increases, these petal-like clusters will also become larger (Figure 1c–e). This change directly affects the catalyst OER activity. Among

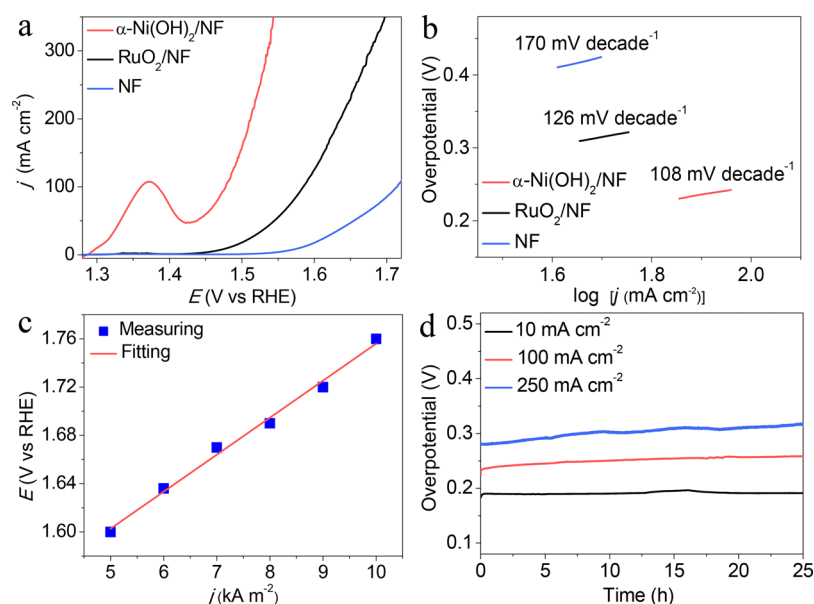


Figure 4. (a) LSV curves and (b) Tafel plots of α -Ni(OH)₂/NF, RuO₂/NF, and NF recorded at a scan rate of 1 mV s⁻¹ in 1.0 M KOH. (c) Test potentials at high current densities for α -Ni(OH)₂/NF in a 1.0 M KOH solution. (d) Galvanostatic electrolysis at 10, 100, and 250 mA cm⁻² of α -Ni(OH)₂/NF in a 1.0 M KOH solution.

them, petal-like clusters deposited at a potential of -2.0 V versus Ag/AgCl have the highest catalytic activity (Figure 2a). The electrodeposition time and electrolyte concentration have a much less direct influence on the morphology of α -Ni(OH)₂, exhibiting a similar petal-like cluster morphology composed of nanosheets (Figures S7 and S8). Because of the optimized electrodeposition conditions, the morphology of α -Ni(OH)₂ prepared by electrodeposition is different from those of other reported α -Ni(OH)₂ by electrodeposition.^{44–47} This unique structure is conducive to the full contact of the catalyst with the electrolyte and the release of gas during the catalytic process. SEM and corresponding elemental mapping images confirm the uniform distribution of Ni and O elements in α -Ni(OH)₂ (Figure S9). TEM images also demonstrate that the 3D petal-like clusters are assembled by the ultrathin nanosheets (Figure 3b). The high-resolution transmission electron microscopy (HRTEM) image shows that the nanosheets have legible lattice fringes with a spacing of ~ 0.235 nm, which is consistent with the (015) crystal plane of α -Ni(OH)₂ (Figure 3d).⁴⁸ Atomic force microscopy (AFM) was carried out to determine the thickness of α -Ni(OH)₂ nanosheets constructed under different conditions (Figure 3c and Figures S10–S12). Interestingly, we found that the electrodeposition potential can affect the thickness of α -Ni(OH)₂ nanosheets. As shown in Figure 2d, the thickness of α -Ni(OH)₂ decreases from -1.5 to -2.0 V versus Ag/AgCl and then gradually increases from -2.0 to -3.0 V versus Ag/AgCl, being consistent with the order of OER overpotential. It is worth noting that the thickness of α -Ni(OH)₂ nanosheets prepared by electrodeposition at -2.0 V versus Ag/AgCl is only approximately 3.0 nm (Figures 2d and 3c and Figure S10b), which is remarkably smaller than most nickel hydroxides reported previously.^{49–51} The 2D ultrathin nanosheet structure endows the catalyst with a large specific surface area and highly exposed catalytic active sites, which should promote its electrocatalytic performance.²⁵ The concentration of Ni(NO₃)₂·6H₂O also has a certain effect on the thickness of α -Ni(OH)₂ nanosheets. As shown in Figure 2e, the thickness of α -Ni(OH)₂ decreases from 0.1 to

0.2 M and increases slightly from 0.2 to 0.4 M. On the contrary, the electrodeposition time has a very slight effect on the thickness of α -Ni(OH)₂ nanosheets, implying that increasing the electrodeposition time mainly increases the number of nanosheets (Figure 2f).

To further reveal the root cause of the effect of electrodeposition conditions on OER activity, we tested the electrochemically active specific surface area (ECSA) and conductivity of these self-supporting electrodes. The ECSA was evaluated by measuring the electrochemical double-layer capacitance (C_{dl}) (Figures S13–S15). The larger the C_{dl} , the greater the ECSA of the catalyst.¹⁰ As shown in Figure S16, the ECSA of the self-supporting electrode increases with the loading mass of α -Ni(OH)₂ and reaches the maximum when the loading mass reaches ~ 9.6 mg cm⁻², and continuing to increase the loading will decrease the ECSA. In particular, the thinner the α -Ni(OH)₂ nanosheets, the larger the ECSA of the catalyst (Figure 2d–f), indicating that the preparation of ultrathin nanosheets is the essential reason for improving the OER activity of the catalyst. High electronic conductivity is also beneficial for enhancing the electrocatalytic activity of the catalyst for OER. EIS was utilized to reveal the charge transfer resistance (R_{CT}) of α -Ni(OH)₂/NF. As shown in Figure S17, an appropriate increase in the catalyst loading mass can significantly improve the electron conductivity, but excessive loading will hinder electron conduction. Generally, on one hand, excessive loading increases the thickness of the nanosheets, making the band gap of the material larger.^{52,53} On the other hand, because nickel hydroxide is far less conductive than metallic nickel, when the loading is excessive, the electron transfer efficiency should be naturally lower.⁴¹ Therefore, we can conclude that accurately regulating the electrodeposition conditions can significantly adjust the ECSA and conductivity of α -Ni(OH)₂/NF to finely tune its catalytic activity.

The intensive electrochemical performance of the optimal α -Ni(OH)₂/NF self-supporting electrode for OER was estimated in 1.0 M KOH through a three-electrode test system. For

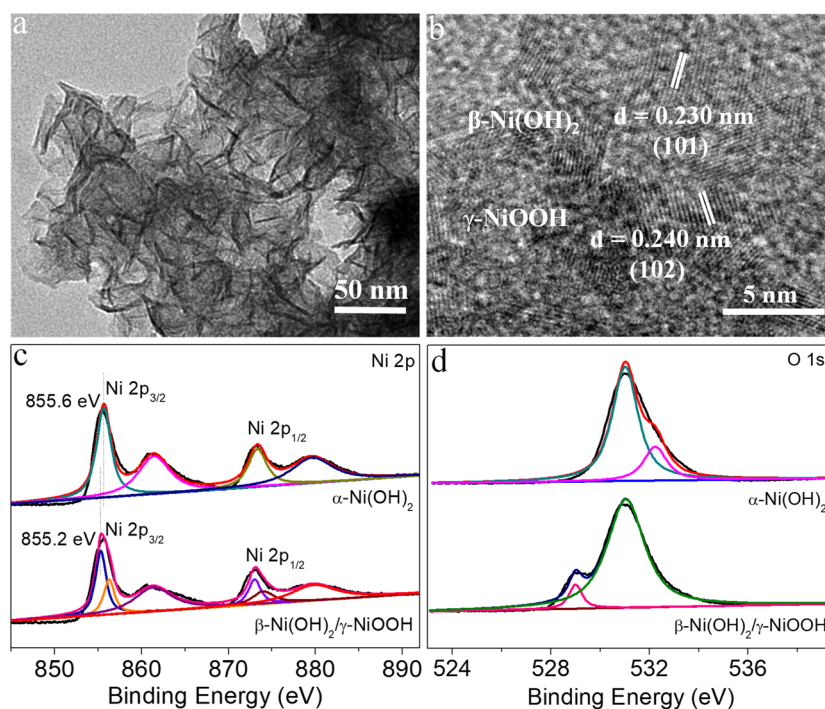


Figure 5. (a) TEM and (b) HRTEM images of α -Ni(OH)₂ after OER. High-resolution XPS spectra of (c) Ni 2p and (d) O 1s of α -Ni(OH)₂ before (up) and after (down) OER. The black curves in panels c and d are the original data.

comparison, we loaded commercial RuO₂ with the same mass on NF and measured its OER activity under the same conditions. All electrodes were activated to a stable state through cyclic voltammetry (CV) before their electrochemical performance was tested. All recorded electrochemical data are the result of *iR* compensation. As depicted in the LSV curves (Figure 4a), the α -Ni(OH)₂/NF self-standing electrode exhibits excellent electrocatalytic activity for OER, requiring a low overpotential of 240 mV to yield a current density of 100 mA cm⁻². In stark contrast, the RuO₂/NF and NF electrodes require overpotentials of 358 and 482 mV, respectively, to yield the same current density. The OER performance of the as-prepared α -Ni(OH)₂/NF is better than those of almost all currently reported nickel hydroxide catalysts (Table S1) and is comparable to those of state-of-the-art OER catalysts.^{54–56} In particular, α -Ni(OH)₂/NF merely demands a quite low overpotential of 290 mV to yield a large current density of 250 mA cm⁻², which is much smaller than that of the benchmark RuO₂/NF (430 mV). We also measured the drive potentials of the α -Ni(OH)₂/NF electrode at higher current densities (500–1000 mA cm⁻²), and the results showed that α -Ni(OH)₂/NF also exhibited good catalytic activity (Figure 4c), reflecting its potential industrial application value.⁵⁷ In addition, the oxidation peak at around 1.37 V versus a RHE is attributed to the conversion of Ni²⁺ to Ni³⁺.²⁷ To highlight the performance of the ultrathin nickel hydroxide nanosheets, we compared the OER catalytic activity of nickel hydroxide nanosheets with different thicknesses obtained under different electrodeposition conditions. The LSV results revealed that the thinner the nickel hydroxide nanosheets, the better their catalytic performance (Figure S1). The OER kinetics of these electrodes were tested through corresponding Tafel plots derived from the LSV curves, which can evaluate the intrinsic electrocatalytic activity. As shown in Figure 4b, the Tafel slope of the α -Ni(OH)₂/NF electrode is 108 mV decade⁻¹, being

much lower than those of RuO₂/NF (126 mV decade⁻¹) and NF (170 mV decade⁻¹), suggesting fast catalytic kinetics of α -Ni(OH)₂/NF for OER. The TOF value of α -Ni(OH)₂/NF is 0.010 s⁻¹ at an overpotential of 300 mV. These results clearly confirm that the α -Ni(OH)₂/NF self-supporting electrode with optimal morphology and loading mass has outstanding electrocatalytic activity for OER.

The ECSA and conductivity of these electrodes were further investigated to analyze the origin of the intrinsic activity. As shown in Figure S18, the *C*_{dl} value of α -Ni(OH)₂/NF is 7.52 mF cm⁻², which is much higher than those of RuO₂/NF (5.88 mF cm⁻²) and NF (1.45 mF cm⁻²). This result suggests that the α -Ni(OH)₂/NF electrode has a higher ECSA and can provide abundant electrocatalytic active sites to greatly promote its OER activity. In addition, as displayed in Figure S19, the *R*_{CT} of α -Ni(OH)₂/NF is smaller than those of RuO₂/NF and NF, demonstrating its excellent electronic conductivity, thereby further boosting its OER performance.

The high stability of the electrocatalyst is also particularly important for industrial applications. Galvanostatic electrolysis measurements were employed to evaluate the long-term stability of α -Ni(OH)₂/NF. When the α -Ni(OH)₂/NF electrode is operated at a current density of 10 mA cm⁻² for 25 h, there is almost no change in the overpotential, demonstrating its excellent stability for OER in a strong alkaline electrolyte. Impressively, α -Ni(OH)₂/NF still maintains remarkable stability at high current densities of 100 and 250 mA cm⁻² (Figure 4d), and the corresponding overpotentials are increased by only 6 and 10 mV, respectively, after electrolysis for 25 h. Moreover, the stability of α -Ni(OH)₂/NF was further evaluated through the LSV measurement after galvanostatic electrolysis at a current density of 10 mA cm⁻². As shown in Figure S20, there is almost no obvious change in the LSV curves before and after electrolysis, confirming the outstanding long-term stability of the α -

Ni(OH)₂/NF self-supporting electrode. Furthermore, α -Ni(OH)₂/NF exhibits an almost 100% Faradaic efficiency for OER (Figure S21).

Studies have shown that the active species of some electrocatalysts will change during the OER process.²⁶ To reveal the real active species for OER, we characterized the composition, morphology, and crystal phase structure of the catalyst after OER. The PXRD pattern of α -Ni(OH)₂ after OER is presented in Figure 3a. The diffraction peaks at 19.25°, 33.07°, 38.54°, 52.10°, 59.05°, and 62.73° can be attributed to the (001), (100), (101), (102), (110), and (111) crystal planes, respectively, of the Ni(OH)₂ phase (JCPDS Card No. 14-0117), and the diffraction peaks at 12.81°, 37.93°, 51.59°, and 66.23° can be attributed to the (003), (102), (108), and (110) crystal planes, respectively, of the NiOOH phase (JCPDS Card No. 06-0075), suggesting that α -Ni(OH)₂ was converted to β -Ni(OH)₂ and γ -NiOOH [β -Ni(OH)₂/ γ -NiOOH] after OER.^{58,59} In addition, we also tested the PXRD of the catalyst after galvanostatic electrolysis at 10 mA cm⁻² for 75 h and found that the two-phase structure [β -Ni(OH)₂/ γ -NiOOH] after the initial transformation can still be maintained (Figure S22). The results confirmed that the in situ-generated β -Ni(OH)₂/ γ -NiOOH was very stable during OER. The SEM and TEM images show that the in situ-transformed β -Ni(OH)₂/ γ -NiOOH maintains the nanosheet morphology (Figure 5a and Figure S23), confirming its remarkable morphological stability during the OER process. Moreover, HRTEM also demonstrated that the crystal structure of α -Ni(OH)₂ had changed. As shown in Figure 5b, the clear lattice fringes with lattice spacings of 0.230 and 0.240 nm correspond to the crystal planes of (101) of β -Ni(OH)₂ and (102) of γ -NiOOH, respectively.⁵⁸ Furthermore, XPS was utilized to reveal the surface chemical composition and element valence state of α -Ni(OH)₂ before and after OER. As shown in Figure 5c, the high-resolution Ni 2p XPS spectrum in α -Ni(OH)₂ before OER exhibits four characteristic peaks, where the two peaks at 855.6 and 873.4 eV correspond to the Ni 2p_{3/2} and Ni 2p_{1/2} features of Ni²⁺, respectively.^{60,61} The two peaks located at 861.4 and 879.6 eV are attributed to the satellite peaks of Ni²⁺.⁵⁸ As shown in Figure 5d, the two characteristic peaks at 531.0 and 532.3 eV can be fitted from the high-resolution O 1s XPS spectrum. Obviously, the characteristic peak located at 530.9 eV can be assigned to the Ni hydroxide species in the α -Ni(OH)₂ nanosheets, and the other characteristic peak at 532.1 eV is associated with water molecules adsorbed on the surface of α -Ni(OH)₂.^{58,60,61} The analysis presented above confirms that the surface of α -Ni(OH)₂ before OER is composed of Ni²⁺ and hydroxide species. After OER, in the high-resolution Ni 2p spectrum, there are two spin orbit doublets and two corresponding satellite peaks (Figure 5c). The first doublet at 855.2 and 873.0 eV can be assigned to Ni²⁺ in β -Ni(OH)₂, which is negatively shifted by ~0.4 eV compared to that of α -Ni(OH)₂.^{58,62} The second one at 856.3 and 874.1 eV corresponds to Ni³⁺, implying the generation of NiOOH.^{58,62,63} From the high-resolution O 1s spectrum (Figure 5d), the two characteristic peaks at 529.1 and 531.0 eV can be fitted, corresponding to the nickel–oxygen (Ni–O) and the protonated oxygen (H–O), respectively.^{58,61–63} The XPS analysis after OER demonstrates that a portion of Ni(OH)₂ was in situ transformed into NiOOH, which has been generally considered to be the truly active species of OER. Many studies have shown that the γ -NiOOH transformed from Ni-based

precatalysts has many defect sites, such as vacancies, edges, corners, stacking faults, etc., which can act as highly active species to enhance OER catalysis.^{22,64–66} Therefore, this phase transition is beneficial for enhancing the catalytic activity of the catalyst. On one hand, α -Ni(OH)₂ is unstable under strong alkaline solutions and can transform easily to the more stable β -Ni(OH)₂; on the other hand, part of the nickel hydroxide will be converted into stable γ -NiOOH active species during the OER process.^{50,64–66} Therefore, this phase transition is also beneficial for enhancing the catalytic stability of the catalyst. In addition, the PDOS calculated by density functional theory (DFT) showed that the d orbital center of the Ni atom in ultrathin NiOOH nanosheets moves toward the Fermi level (that is, toward the higher energy level) compared with the bulk state (Figure S24). This is beneficial for enhancing the adsorption of OER intermediates on Ni atoms (such as O*), thereby improving the catalytic activity.^{66–68}

To further prove the universality of this electrodeposition method for substrate materials, we also electrodeposited ultrathin nickel hydroxide nanosheets on copper foam (CF) and carbon cloth (CC) conductive substrates under the same conditions (Figure S25). As shown in Figures S26–S28, the crystal structure and morphology of nickel hydroxide electrodeposited on CF and CC are the same as those on NF. The as-obtained α -Ni(OH)₂/CF and α -Ni(OH)₂/CC self-standing electrodes also exhibit outstanding electrocatalytic activity and stability for OER. LSV curves reveal that α -Ni(OH)₂/CF and α -Ni(OH)₂/CC require overpotentials of 278 and 375 mV, respectively, to yield a large current density of 100 mA cm⁻² (Figure S29). As shown in Figure S30, the Tafel slopes of the α -Ni(OH)₂/CF and α -Ni(OH)₂/CC electrode are 117 and 135 mV decade⁻¹, respectively, suggesting their good catalytic kinetics for OER. To avoid the interference from the oxidation peaks of Ni²⁺ to Ni³⁺, galvanostatic electrolysis measurements were carried out to determine the overpotentials at the current density of 10 mA cm⁻². As shown in Figure S31, α -Ni(OH)₂/CF and α -Ni(OH)₂/CC merely demand overpotentials of 208 and 240 mV, respectively, to yield a current density of 10 mA cm⁻². The results show that the OER activity of α -Ni(OH)₂/CF is comparable to that of α -Ni(OH)₂/NF. Moreover, the α -Ni(OH)₂/CF and α -Ni(OH)₂/CC self-standing electrodes also exhibit remarkable long-term stability at current densities of 10 and 100 mA cm⁻² (Figure S31). The experimental results presented above confirm that electrodeposited α -Ni(OH)₂ species on different conductive substrates all exhibit high electrocatalytic activity for OER, which can be attributed to the large active specific surface area as well as highly exposed catalytic active sites of the ultrathin nanosheets.

4. CONCLUSIONS

In summary, we have fabricated ultrathin nickel hydroxide nanosheets through a facile one-step electrodeposition method. The micro–nano structure as well as the active specific surface area and electron conductivity of the α -Ni(OH)₂/NF self-supporting electrode can be optimized by adjusting the electrodeposition conditions, especially the electrodeposition potential, thereby greatly promoting its electrocatalytic activity for OER. Impressively, the α -Ni(OH)₂/NF self-standing electrode exhibits excellent electrocatalytic performance for OER, requiring an extremely low overpotential of 192 mV to yield a current density of 10 mA cm⁻² in 1.0 M KOH, being one of the best OER catalysts reported to date. In particular, α -Ni(OH)₂/NF requires low

overpotentials of only 240 and 290 mV to yield large current densities of 100 and 250 mA cm⁻² in 1.0 M KOH, respectively. Moreover, α -Ni(OH)₂/NF shows great long-term stability at both low and high current densities. The excellent electrocatalytic performance of α -Ni(OH)₂/NF can be attributed to the following aspects. (1) The 3D petal-like clusters constructed from ultrathin nanosheets can expose abundant catalytically active sites. (2) The optimization of the loading mass of α -Ni(OH)₂ increases the specific surface area and conductivity of the self-supporting electrode. (3) The in situ-transformed NiOOH active species can further boost the electrocatalytic activity. (4) The direct growth of the catalyst on the conductive substrate is conducive to promoting electron conduction, enhancing mass transmission, and stabilizing the working electrode. These results highlight the importance of electrodeposition parameters on electrochemical material synthesis and provide a new idea for the controllable electrochemical preparation of electrocatalysts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.inorgchem.0c03771>.

SEM images, TEM images, EDS elemental mapping images, AFM images, and XRD patterns of the products; supplemental data for electrocatalytic performances; DFT calculation results; and faradic efficiency (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Chun-Ting He – Key Laboratory of Functional Small Organic Molecule, Ministry of Education, College of Chemistry and Chemical Engineering, Jiangxi Normal University, Nanchang 330022, China; orcid.org/0000-0003-0009-7693; Email: hct@jxnu.edu.cn

Authors

Li-Ming Cao – Key Laboratory of Functional Small Organic Molecule, Ministry of Education, College of Chemistry and Chemical Engineering, Jiangxi Normal University, Nanchang 330022, China

Qing-Cai Cao – Key Laboratory of Functional Small Organic Molecule, Ministry of Education, College of Chemistry and Chemical Engineering, Jiangxi Normal University, Nanchang 330022, China

Jia Zhang – College of Life Science, Jiangxi Normal University, Nanchang 330022, China

Xuan-Yi Zhu – Key Laboratory of Functional Small Organic Molecule, Ministry of Education, College of Chemistry and Chemical Engineering, Jiangxi Normal University, Nanchang 330022, China

Rong-Zhi Sun – Key Laboratory of Functional Small Organic Molecule, Ministry of Education, College of Chemistry and Chemical Engineering, Jiangxi Normal University, Nanchang 330022, China

Zi-Yi Du – Key Laboratory of Functional Small Organic Molecule, Ministry of Education, College of Chemistry and Chemical Engineering, Jiangxi Normal University, Nanchang 330022, China

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.inorgchem.0c03771>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (21901088, 21901089, and 21971091) and the Natural Science Foundation of Jiangxi Province (20192BAB213003, 20202ZDB01004, and 20192ACB20013). C.-T.H. acknowledges support of the “Young Elite Scientists Sponsorship Program” by CAST.

■ REFERENCES

- (1) Yuan, T. W.; Li, Z. J.; Zhang, W. S.; Xue, Z. G.; Wang, X. Q.; Ma, Z. H.; Fan, Y.; Xu, J. Q.; Wu, Y. E. Highly Sensitive Ethanol Gas Sensor Based on Ultrathin Nanosheets Assembled Bi₂WO₆ with Composite Phase. *Sci. Bull.* **2019**, *64*, 595–602.
- (2) Xiong, C.; Tian, L.; Xiao, C. C.; Xue, Z. G.; Zhou, F. Y.; Zhou, H.; Zhao, Y. F.; Chen, M.; Wang, Q. P.; Qu, Y. T.; Hu, Y. D.; Wang, W. Y.; Zhang, Y.; Zhou, X.; Wang, Z. Y.; Yin, P. Q.; Mao, Y.; Yu, Z. Q.; Cao, Y. Q.; Duan, X. Z.; Zheng, L.; Wu, Y. Construction of Highly Accessible Single Co Site Catalyst for Glucose Detection. *Sci. Bull.* **2020**, *65*, 2100–2106.
- (3) Wan, J. W.; Zhao, Z. H.; Shang, H. S.; Peng, B.; Chen, W. X.; Pei, J. J.; Zheng, L. R.; Dong, J. C.; Cao, R.; Sarangi, R.; Jiang, Z. L.; Zhou, D. N.; Zhuang, Z. B.; Zhang, J. T.; Wang, D. S.; Li, Y. D. In Situ Phosphatizing of Triphenylphosphine Encapsulated within Metal-Organic Frameworks to Design Atomic Co1-PIN3 Interfacial Structure for Promoting Catalytic Performance. *J. Am. Chem. Soc.* **2020**, *142*, 8431–8439.
- (4) Yun, Q. B.; Lu, Q. P.; Zhang, X.; Tan, C. L.; Zhang, H. Three-Dimensional Architectures Constructed from Transition-Metal Dichalcogenide Nanomaterials for Electrochemical Energy Storage and Conversion. *Angew. Chem., Int. Ed.* **2018**, *57*, 626–646.
- (5) Dou, Y. H.; He, C. T.; Zhang, L.; Yin, H. J.; Al-Mamun, M.; Ma, J. M.; Zhao, H. J. Approaching the Activity Limit of CoSe₂ for Oxygen Evolution via Fe Doping and Co Vacancy. *Nat. Commun.* **2020**, *11*, 1664.
- (6) Jadhav, R. G.; Singh, D.; Krivoschapkin, P. V.; Das, A. K. Electrodeposited Organic-Inorganic Nanohybrid as Robust Bifunctional Electrocatalyst for Water Splitting. *Inorg. Chem.* **2020**, *59*, 7469–7478.
- (7) Xu, Y. T.; Xiao, X. F.; Ye, Z. M.; Zhao, S. L.; Shen, R. G.; He, C. T.; Zhang, J. P.; Li, Y. D.; Chen, X. M. Cage-Confinement Pyrolysis Route to Ultrasmall Tungsten Carbide Nanoparticles for Efficient Electrocatalytic Hydrogen Evolution. *J. Am. Chem. Soc.* **2017**, *139*, 5285–5288.
- (8) Yu, Y. F.; Shi, Y. M.; Zhang, B. Synergetic Transformation of Solid Inorganic-Organic Hybrids into Advanced Nanomaterials for Catalytic Water Splitting. *Acc. Chem. Res.* **2018**, *51*, 1711–1721.
- (9) Jin, H. Y.; Guo, C. X.; Liu, X.; Liu, J. L.; Vasileff, A.; Jiao, Y.; Zheng, Y.; Qiao, S. Z. Emerging Two-Dimensional Nanomaterials for Electrocatalysis. *Chem. Rev.* **2018**, *118*, 6337–6408.
- (10) Cao, L. M.; Lu, D.; Zhong, D. C.; Lu, T. B. Prussian Blue Analogues and Their Derived Nanomaterials for Electrocatalytic Water Splitting. *Coord. Chem. Rev.* **2020**, *407*, 213156.
- (11) Wang, Y. O.; Vogel, A.; Sachs, M.; Sprick, R. S.; Wilbraham, L.; Moniz, S. J. A.; Godin, R.; Zwienerburg, M. A.; Durrant, J. R.; Cooper, A. I.; Tang, J. W. Current Understanding and Challenges of Solar-Driven Hydrogen Generation Using Polymeric Photocatalysts. *Nat. Energy* **2019**, *4*, 746–760.
- (12) Zhao, J.; Rong, F.; Yao, Y.; Fan, W. J.; Li, M. R.; Yang, Q. H. Co NP/NC Hollow Nanoparticles Derived from Yolk-Shell Structured ZIFs@Polydopamine as Bifunctional Electrocatalysts for Water Oxidation and Oxygen Reduction Reactions. *J. Energy Chem.* **2018**, *27*, 1261–1267.
- (13) Wang, Y.; Zhang, W. H.; Deng, D. H.; Bao, X. H. Two-Dimensional Materials Confining Single Atoms for Catalysis. *Chin. J. Catal.* **2017**, *38*, 1443–1453.

- (14) Gao, Z.; Gong, L.; He, X. Q.; Su, X. M.; Xiao, L. H.; Luo, F. General Strategy to Fabricate Metal-Incorporated Pyrolysis-Free Covalent Organic Framework for Efficient Oxygen Evolution Reaction. *Inorg. Chem.* **2020**, *59*, 4995–5003.
- (15) Zhang, B.; Zheng, X. L.; Voznyy, O.; Comin, R.; Bajdich, M.; Garcia-Melchor, M.; Han, L. L.; Xu, J. X.; Liu, M.; Zheng, L. R.; Garcia de Arquer, F. P.; Dinh, C. T.; Fan, F. J.; Yuan, M. J.; Yassitepe, E.; Chen, N.; Regier, T.; Liu, P. F.; Li, Y. H.; De Luna, P.; Janmohamed, A.; Xin, H. L. L.; Yang, H. G.; Vojvodic, A.; Sargent, E. H. Homogeneously Dispersed Multimetal Oxygen-Evolving Catalysts. *Science* **2016**, *352*, 333–337.
- (16) Zhang, D.; Peng, L. X.; Yang, Z. G.; Yang, Y.; Li, H. P. Gold-Supported Nanostructured NiFeCoPr Hydroxide as a High-Performance Supercapacitor Electrode and Electrocatalyst toward the Oxygen Evolution Reaction. *Inorg. Chem.* **2019**, *58*, 15841–15852.
- (17) Yao, Y. C.; Hu, S. L.; Chen, W. X.; Huang, Z. Q.; Wei, W. C.; Yao, T.; Liu, R. R.; Zang, K. T.; Wang, X. Q.; Wu, G.; Yuan, W. J.; Yuan, T. W.; Zhu, B. Q.; Liu, W.; Li, Z. J.; He, D. S.; Xue, Z. G.; Wang, Y.; Zheng, X. S.; Dong, J. C.; Chang, C. R.; Chen, Y. X.; Hong, X.; Luo, J.; Wei, S. Q.; Li, W. X.; Strasser, P.; Wu, Y. E.; Li, Y. D. Engineering The Electronic Structure of Single Atom Ru Sites via Compressive Strain Boosts Acidic Water Oxidation Electrocatalysis. *Nat. Catal.* **2019**, *2*, 304–313.
- (18) Xu, J.; Wang, M. S.; Yang, F.; Ju, X. Q.; Jia, X. L. Self-Supported Porous Ni-Fe-W Hydroxide Nanosheets on Carbon Fiber: A Highly Efficient Electrode for Oxygen Evolution Reaction. *Inorg. Chem.* **2019**, *58*, 13037–13048.
- (19) Lee, Y.; Suntivich, J.; May, K. J.; Perry, E. E.; Shao-Horn, Y. Synthesis and Activities of Rutile IrO₂ and RuO₂ Nanoparticles for Oxygen Evolution in Acid and Alkaline Solutions. *J. Phys. Chem. Lett.* **2012**, *3*, 399–404.
- (20) Gorlin, M.; Ferreira de Araujo, J.; Schmies, H.; Bernsmeier, D.; Dresp, S.; Gliech, M.; Jusys, Z.; Cherev, P.; Kraehnert, R.; Dau, H.; Strasser, P. Tracking Catalyst Redox States and Reaction Dynamics in Ni-Fe Oxyhydroxide Oxygen Evolution Reaction Electrocatalysts: The Role of Catalyst Support and Electrolyte pH. *J. Am. Chem. Soc.* **2017**, *139*, 2070–2082.
- (21) Zhang, J. T.; Yu, L.; Chen, Y.; Lu, X. F.; Gao, S. Y.; Lou, X. W. Designed Formation of Double-Shelled Ni-Fe Layered-Double-Hydroxide Nanocages for Efficient Oxygen Evolution Reaction. *Adv. Mater.* **2020**, *32*, 1906432.
- (22) Gao, M. R.; Sheng, W. C.; Zhuang, Z. B.; Fang, Q. R.; Gu, S.; Jiang, J.; Yan, Y. S. Efficient Water Oxidation Using Nanostructured α -Nickel-Hydroxide as an Electrocatalyst. *J. Am. Chem. Soc.* **2014**, *136*, 7077–7084.
- (23) Xu, H.; Shang, H. Y.; Di, J. W.; Du, Y. K. Geometric and Electronic Engineering of Mn-Doped Cu(OH)(2) Hexagonal Nanorings for Superior Oxygen Evolution Reaction Electrocatalysis. *Inorg. Chem.* **2019**, *58*, 15433–15442.
- (24) Wang, Y. Y.; Xie, C.; Zhang, Z. Y.; Liu, D. D.; Chen, R.; Wang, S. Y. In Situ Exfoliated, N-Doped, and Edge-Rich Ultrathin Layered Double Hydroxides Nanosheets for Oxygen Evolution Reaction. *Adv. Funct. Mater.* **2018**, *28*, 1703363.
- (25) Gao, M. R.; Cao, X.; Gao, Q.; Xu, Y. F.; Zheng, Y. R.; Jiang, J.; Yu, S. H. Nitrogen-Doped Graphene Supported CoSe₂ Nanobelt Composite Catalyst for Efficient Water Oxidation. *ACS Nano* **2014**, *8*, 3970–3978.
- (26) Wang, J. H.; Cui, W.; Liu, Q.; Xing, Z. C.; Asiri, A. M.; Sun, X. P. Recent Progress in Cobalt-Based Heterogeneous Catalysts for Electrochemical Water Splitting. *Adv. Mater.* **2016**, *28*, 215–230.
- (27) Vij, V.; Sultan, S.; Harzandi, A. M.; Meena, A.; Tiwari, J. N.; Lee, W. G.; Yoon, T.; Kim, K. S. Nickel-Based Electrocatalysts for Energy-Related Applications: Oxygen Reduction, Oxygen Evolution, and Hydrogen Evolution Reactions. *ACS Catal.* **2017**, *7*, 7196–7225.
- (28) Hong, Y. R.; Kim, K. M.; Ryu, J. H.; Mhin, S.; Kim, J.; Ali, G.; Chung, K. Y.; Kang, S.; Han, H. Dual-Phase Engineering of Nickel Boride-Hydroxide Nanoparticles toward High-Performance Water Oxidation Electrocatalysts. *Adv. Funct. Mater.* **2020**, *30*, 2004330.
- (29) Harvey, A.; He, X. Y.; Godwin, I. J.; Backes, C.; McAteer, D.; Berner, N. C.; McEvoy, N.; Ferguson, A.; Shmeliov, A.; Lyons, M. E. G.; Nicolosi, V.; Duesberg, G. S.; Donegan, J. F.; Coleman, J. N. Production of Ni(OH)₂ Nanosheets by Liquid Phase Exfoliation: from Optical Properties to Electrochemical Applications. *J. Mater. Chem. A* **2016**, *4*, 11046–11059.
- (30) Xiong, D. H.; Li, W.; Liu, L. F. Vertically Aligned Porous Nickel(II) Hydroxide Nanosheets Supported on Carbon Paper with Long-Term Oxygen Evolution Performance. *Chem. - Asian J.* **2017**, *12*, 543–551.
- (31) Zhang, Q.; Zhang, C. C.; Liang, J. B.; Yin, P. G.; Tian, Y. Orthorhombic α -NiOOH Nanosheet Arrays: Phase Conversion and Efficient Bifunctional Electrocatalysts for Full Water Splitting. *ACS Sustainable Chem. Eng.* **2017**, *5*, 3808–3818.
- (32) Gao, M.; He, L.; Guo, Z. Y.; Yuan, Y. R.; Li, W. W. Sulfate-Functionalized Nickel Hydroxide Nanobelts for Sustained Oxygen Evolution. *ACS Appl. Mater. Interfaces* **2020**, *12*, 443–450.
- (33) Dou, Y. H.; Zhang, L.; Xu, J. T.; He, C. T.; Xu, X.; Sun, Z. Q.; Liao, T.; Nagy, B.; Liu, P. R.; Dou, S. X. Manipulating the Architecture of Atomically Thin Transition Metal (Hydr)oxides for Enhanced Oxygen Evolution Catalysis. *ACS Nano* **2018**, *12*, 1878–1886.
- (34) Liu, Y. H.; Gokcen, D.; Bertocci, U.; Moffat, T. P. Self-Terminating Growth of Platinum Films by Electrochemical Deposition. *Science* **2012**, *338*, 1327–1330.
- (35) Mahenderkar, N. K.; Chen, Q. Z.; Liu, Y. C.; Duchild, A. R.; Hofheins, S.; Chason, E.; Switzer, J. A. Epitaxial lift-off of electrodeposited single-crystal gold foils for flexible electronics. *Science* **2017**, *355*, 1203–1206.
- (36) Li, Z. H.; Shao, M. F.; An, H. L.; Wang, Z. X.; Xu, S. M.; Wei, M.; Evans, D. G.; Duan, X. Fast Electrosynthesis of Fe-Containing Layered Double Hydroxide Arrays toward Highly Efficient Electrocatalytic Oxidation Reactions. *Chem. Sci.* **2015**, *6*, 6624–6631.
- (37) Zhang, W.; Wu, Y. Z.; Qi, J.; Chen, M. X.; Cao, R. A Thin NiFe Hydroxide Film Formed by Stepwise Electrodeposition Strategy with Significantly Improved Catalytic Water Oxidation Efficiency. *Adv. Energy Mater.* **2017**, *7*, 1602547.
- (38) Li, X. P.; Wang, Y.; Wang, J. J.; Da, Y. M.; Zhang, J. F.; Li, L. L.; Zhong, C.; Deng, Y. D.; Han, X. P.; Hu, W. B. Sequential Electrodeposition of Bifunctional Catalytically Active Structures in MoO₃/Ni-NiO Composite Electrocatalysts for Selective Hydrogen and Oxygen Evolution. *Adv. Mater.* **2020**, *32*, 2003414.
- (39) Morales-Guio, C. G.; Liardet, L.; Hu, X. L. Oxidatively Electrodeposited Thin-Film Transition Metal (Oxy)hydroxides as Oxygen Evolution Catalysts. *J. Am. Chem. Soc.* **2016**, *138*, 8946–8957.
- (40) Lu, X. Y.; Zhao, C. A. Electrodeposition of Hierarchically Structured Three-Dimensional Nickel-Iron Electrodes for Efficient Oxygen Evolution at High Current Densities. *Nat. Commun.* **2015**, *6*, 6616.
- (41) Yan, Z. H.; Sun, H. M.; Chen, X.; Liu, H. H.; Zhao, Y. R.; Li, H. X.; Xie, W.; Cheng, F. Y.; Chen, J. Anion Insertion Enhanced Electrodeposition of Robust Metal Hydroxide/Oxide Electrodes for Oxygen Evolution. *Nat. Commun.* **2018**, *9*, 2373.
- (42) Hou, J. G.; Wu, Y. Z.; Zhang, B.; Cao, S. Y.; Li, Z. W.; Sun, L. C. Rational Design of Nanoarray Architectures for Electrocatalytic Water Splitting. *Adv. Funct. Mater.* **2019**, *29*, 1808367.
- (43) Cai, G. R.; Zhang, W.; Jiao, L.; Yu, S. H.; Jiang, H. L. Template-Directed Growth of Well-Aligned MOF Arrays and Derived Self-Supporting Electrodes for Water Splitting. *Chem.* **2017**, *2*, 791–802.
- (44) Yao, K. L.; Zhai, M. H.; Ni, Y. H. α -Ni(OH)₂·0.75H₂O Nanofilms on Ni Foam from Simple NiCl₂ Solution: Fast Electrodeposition, Formation Mechanism and Application as an Efficient Bifunctional Electrocatalyst for Overall Water Splitting in Alkaline Solution. *Electrochim. Acta* **2019**, *301*, 87–96.
- (45) Xing, Z. C.; Gan, L. F.; Wang, J.; Yang, X. R. Experimental and Theoretical Insights into Sustained Water Splitting with an Electrodeposited Nanoporous Nickel Hydroxide@Nickel Film as an Electrocatalyst. *J. Mater. Chem. A* **2017**, *5*, 7744–7748.

- (46) Balram, A.; Zhang, H. F.; Santhanagopalan, S. In Situ Decoration of Stainless Steel Nanoparticles for Synergistic Enhancement of α -Ni(OH)₂ Oxygen Evolution Reaction Catalysis. *Mater. Chem. Front.* **2017**, *1*, 2376–2382.
- (47) Wu, J.; Subramaniam, J.; Liu, Y. Q.; Geng, D. S.; Meng, X. B. Facile Assembly of Ni(OH)₂ Nanosheets on Nitrogen-Doped Carbon Nanotubes Network as High-Performance Electrocatalyst for Oxygen Evolution Reaction. *J. Alloys Compd.* **2018**, *731*, 766–773.
- (48) Xie, M. J.; Duan, S. Y.; Shen, Y.; Fang, K.; Wang, Y. Z.; Lin, M.; Guo, X. F. In-Situ-Grown Mg(OH)₂-Derived Hybrid α -Ni(OH)₂ for Highly Stable Supercapacitor. *Acs Energy Lett.* **2016**, *1*, 814–819.
- (49) Mahala, C.; Devi Sharma, M.; Basu, M. Fe-Doped Nickel Hydroxide/Nickel Oxyhydroxide Function as an Efficient Catalyst for the Oxygen Evolution Reaction. *ChemElectroChem* **2019**, *6*, 3488–3498.
- (50) Luan, C. L.; Liu, G. L.; Liu, Y. J.; Yu, L.; Wang, Y.; Xiao, Y.; Qiao, H. Y.; Dai, X. P.; Zhang, X. Structure Effects of 2D Materials on α -Nickel Hydroxide for Oxygen Evolution Reaction. *ACS Nano* **2018**, *12*, 3875–3885.
- (51) Wang, L. L.; Zhang, J.; Jiang, W. T.; Zhao, H.; Liu, H. Z. Free-Standing, Flexible β -Ni(OH)₂/Electrochemically-Exfoliated Graphene Film Electrode for Efficient Oxygen Evolution. *Appl. Surf. Sci.* **2018**, *433*, 88–93.
- (52) Sun, Y. F.; Gao, S.; Lei, F. C.; Xiao, C.; Xie, Y. Ultrathin Two-Dimensional Inorganic Materials: New Opportunities for Solid State Nanochemistry. *Acc. Chem. Res.* **2015**, *48*, 3–12.
- (53) Sun, Y. F.; Gao, S.; Lei, F. C.; Xie, Y. Atomically-Thin Two-Dimensional Sheets for Understanding Active Sites in Catalysis. *Chem. Soc. Rev.* **2015**, *44*, 623–636.
- (54) Kou, T. Y.; Wang, S. W.; Hauser, J. L.; Chen, M. P.; Oliver, S. R. J.; Ye, Y. F.; Guo, J. H.; Li, Y. Ni Foam-Supported Fe-Doped β -Ni(OH)₂ Nanosheets Show Ultralow Overpotential for Oxygen Evolution Reaction. *Acs Energy Lett.* **2019**, *4*, 622–628.
- (55) Rao, Y.; Wang, Y.; Ning, H.; Li, P.; Wu, M. B. Hydrotalcite-like Ni(OH)₂ Nanosheets in Situ Grown on Nickel Foam for Overall Water Splitting. *ACS Appl. Mater. Interfaces* **2016**, *8*, 33601–33607.
- (56) Jiang, J.; Sun, F. F.; Zhou, S.; Hu, W.; Zhang, H.; Dong, J. C.; Jiang, Z.; Zhao, J. J.; Li, J. F.; Yan, W. S.; Wang, M. Atomic-level Insight into Super-Efficient Electrocatalytic Oxygen Evolution on Iron and Vanadium Co-doped Nickel (Oxy)hydroxide. *Nat. Commun.* **2018**, *9*, 2885.
- (57) Liu, T. T.; Gao, W. B.; Wang, Q. Q.; Dou, M. L.; Zhang, Z. P.; Wang, F. Selective Loading of Atomic Platinum on a RuCeOx Support Enables Stable Hydrogen Evolution at High Current Densities. *Angew. Chem., Int. Ed.* **2020**, *59*, 20423–20427.
- (58) Cao, Q. H.; Luo, M.; Huang, Y. T.; Liu, Q.; Kong, X. X.; Lei, J. L.; Jiang, Z.; Wang, J. H. Temperature and Doping-tuned Coordination Environments Around Electroactive Centers in Fe-doped $\alpha(\beta)$ -Ni(OH)₂ for Excellent Water Splitting. *Sustain. Energy Fuels* **2020**, *4*, 1522–1531.
- (59) Yu, X. W.; Zhao, J.; Zheng, L. R.; Tong, Y.; Zhang, M.; Xu, G. C.; Li, C.; Ma, J.; Shi, G. Q. Hydrogen Evolution Reaction in Alkaline Media: Alpha- or Beta-Nickel Hydroxide on the Surface of Platinum? *Acs Energy Lett.* **2018**, *3*, 237–244.
- (60) Bo, X.; Hocking, R. K.; Zhou, S.; Li, Y. B.; Chen, X. J.; Zhuang, J. C.; Du, Y.; Zhao, C. Capturing the active sites of multimetallic (oxy)hydroxides for the oxygen evolution reaction. *Energy Environ. Sci.* **2020**, *13*, 4225–4237.
- (61) Zhu, K. Y.; Liu, H. Y.; Li, M. R.; Li, X. N.; Wang, J. H.; Zhu, X. F.; Yang, W. S. Atomic-scale topochemical preparation of crystalline Fe³⁺-doped β -Ni(OH)₂ for an ultrahighrate oxygen evolution reaction. *J. Mater. Chem. A* **2017**, *5*, 7753–7758.
- (62) Kim, B.; Oh, A.; Kabiraz, M. K.; Hong, Y.; Joo, J.; Baik, H.; Choi, S. I.; Lee, K. NiOOH Exfoliation-Free Nickel Octahedra as Highly Active and Durable Electrocatalysts Toward the Oxygen Evolution Reaction in an Alkaline Electrolyte. *ACS Appl. Mater. Interfaces* **2018**, *10*, 10115–10122.
- (63) Mavric, A.; Fanetti, M.; Lin, Y. T.; Valant, M.; Cui, C. H. Spectroelectrochemical Tracking of Nickel Hydroxide Reveals Its Irreversible Redox States upon Operation at High Current Density. *ACS Catal.* **2020**, *10*, 9451–9457.
- (64) Bediako, D. K.; Surendranath, Y.; Nocera, D. G. Mechanistic Studies of the Oxygen Evolution Reaction Mediated by a Nickel-Borate Thin Film Electrocatalyst. *J. Am. Chem. Soc.* **2013**, *135*, 3662–3674.
- (65) Smith, R. D. L.; Berlinguette, C. P. Accounting for the Dynamic Oxidative Behavior of Nickel Anodes. *J. Am. Chem. Soc.* **2016**, *138*, 1561–1567.
- (66) Trotochaud, L.; Young, S. L.; Ranney, J. K.; Boettcher, S. W. Nickel-Iron Oxyhydroxide Oxygen-Evolution Electrocatalysts: The Role of Intentional and Incidental Iron Incorporation. *J. Am. Chem. Soc.* **2014**, *136*, 6744–6753.
- (67) Shin, H.; Xiao, H.; Goddard, W. A. In Silico Discovery of New Dopants for Fe-Doped Ni Oxyhydroxide (Ni_{1-x}Fe_xOOH) Catalysts for Oxygen Evolution Reaction. *J. Am. Chem. Soc.* **2018**, *140*, 6745–6748.
- (68) Govind Rajan, A.; Martinez, J. M. P.; Carter, E. A. Facet-Independent Oxygen Evolution Activity of Pure β -NiOOH: Different Chemistries Leading to Similar Overpotentials. *J. Am. Chem. Soc.* **2020**, *142*, 3600–3612.